



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

SECR1213 Network Communications

SESSION 25/26, SEMESTER 1

TASK 1: PROJECT SETUP

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Section	01
Group Name	TTL Network Solutions
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1.0 Introduction

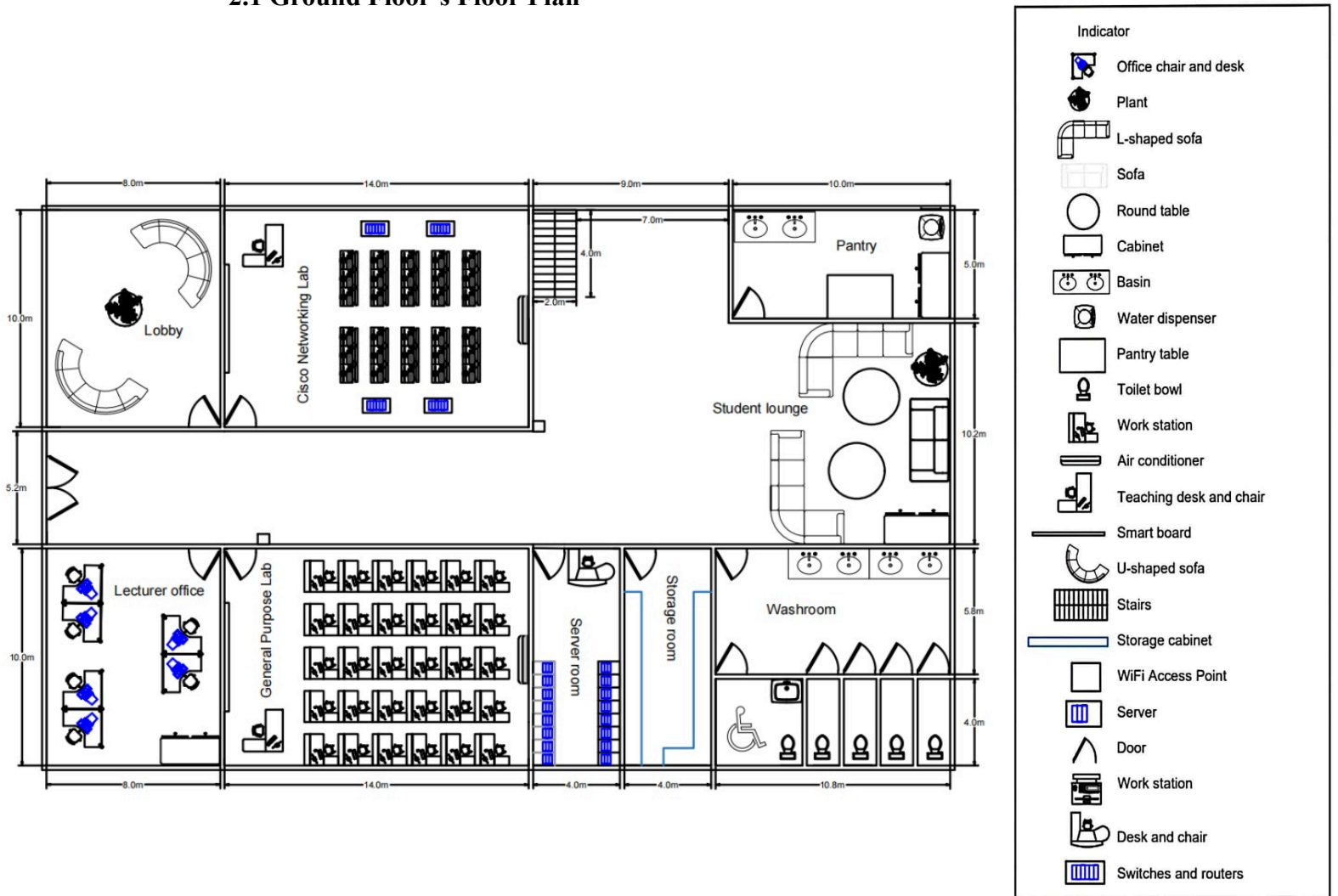
This task focuses on designing a clear, accurate, and functional floor plan for the required building areas, ensuring that each space meets the project requirements and supports seamless network communication. The deliverables include detailed layouts for the ground floor, first floor, Cisco Network Lab, Embedded Lab, two general-purpose labs, hybrid classroom, student lounge, and video conferencing room.

To maintain precision, all floor plans were created using appropriate scale, offering an accurate representation of room sizes, workstation arrangements, and equipment placement. Our group, TTL Network Solutions, collaborated closely to ensure that every design element aligns with the objectives of the project—prioritizing clarity, appropriate labeling, logical flow, and creativity in the layout presentation.

The completed floor plans effectively utilize the available space, accommodate all required equipment including 30 workstations per lab, and adhere to the specifications outlined in the case study. By applying the right tools and a structured planning approach, this task establishes the foundation for efficient network design and successful implementation in the subsequent phases of the project.

2.0 Floor Plan Design

2.1 Ground Floor's Floor Plan



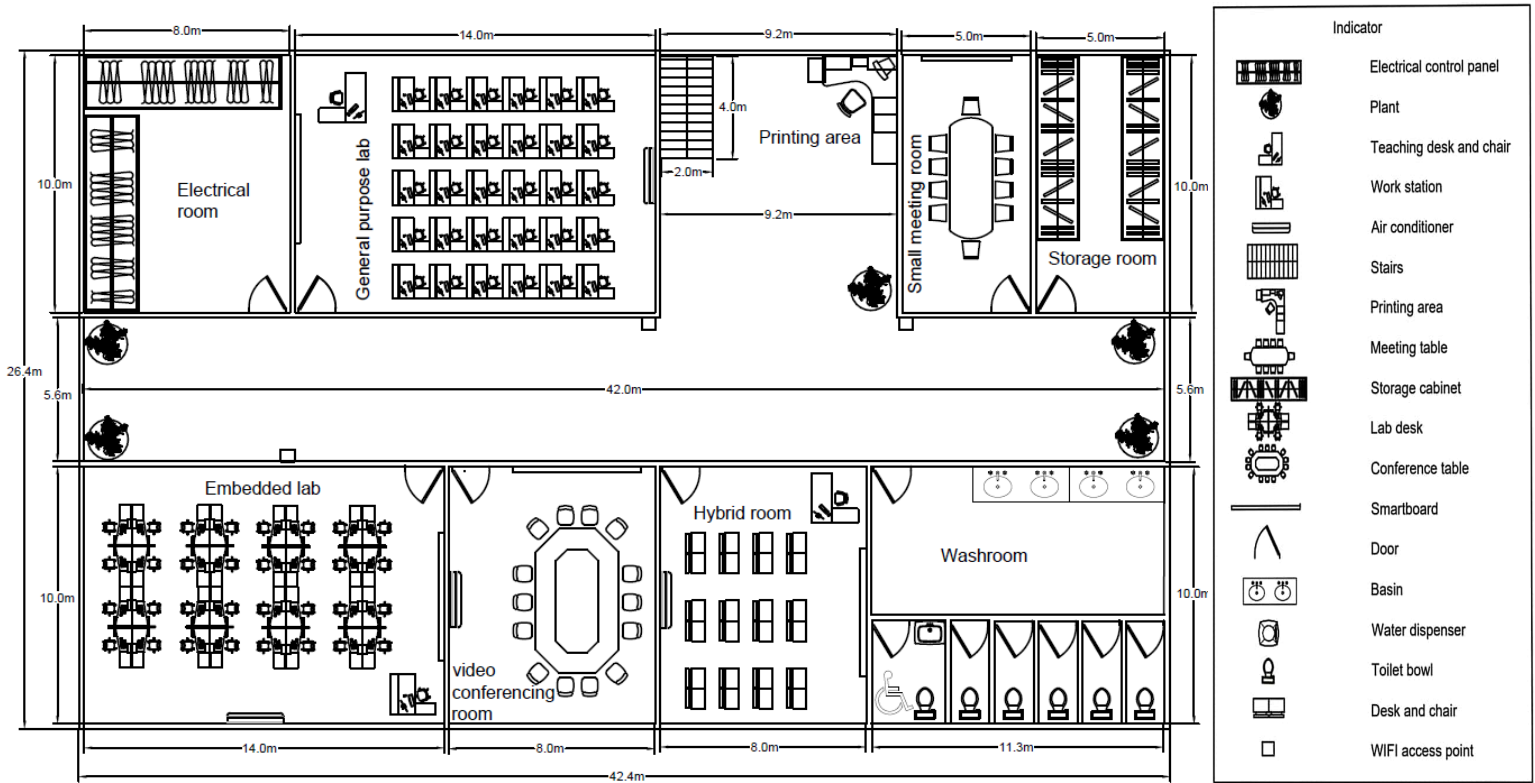
The ground floor plan represents a well-organized educational and training facility designed to support teaching, learning, collaboration, and essential technical operations. To one side of the layout is the Cisco Networking Lab, a specialized training room equipped with multiple workstations, routers, switches, a teaching desk, and a smart board. This lab is designed to support hands-on networking practice and technical instruction. Nearby is the General Purpose Lab, which serves as a flexible computer or project room. It features standard workstations that can accommodate classes, assignments, and general computing tasks.

The plan also includes a Lecturer Office, providing a private and comfortable workspace furnished with a desk, chair, cabinet, and decorative plant. This room offers privacy and a professional environment for academic work or consultations. Adjacent to the teaching and working areas is the Student Lounge, a dedicated rest and social zone equipped with an L-shaped

sofa, U-shaped sofa, round tables, and soft seating, giving students a comfortable place to relax, study informally, or interact with peers. Supporting facilities include a compact Pantry with a table and water dispenser for casual refreshment, as well as a Washroom fitted with a basin and toilet bowl for users' convenience.

On the service side of the layout, the Storage Room provides secure space for equipment, materials, and teaching supplies, while the Server Room houses essential technical infrastructure such as servers, networking switches, routers, and a WiFi access point. This room is designed for controlled access, ensuring proper maintenance and protection of sensitive equipment. A staircase is also included, offering vertical circulation to upper floors without interfering with the functionality of the main rooms.

2.2 First Floor's Floor Plan



This floor plan presents a well-organized and functional layout designed to support technical education, collaborative work, and essential facility operations. On the upper portion of the plan, the Electrical Room is positioned on the left, containing electrical control panels arranged neatly for maintenance and infrastructure support. Adjacent to it is the General Purpose Lab, a spacious classroom equipped with rows of workstations and a teaching desk, making it suitable for theory-based lessons, computer work, or general academic sessions. A centrally located Printing Area sits nearby, easily accessible to students and staff without interfering with classroom activities. Toward the right side, a Small Meeting Room is included for discussions, group projects, or staff consultations, while the Storage Room provides ample cabinet space to organize materials, equipment, and supplies, ensuring the rest of the learning environment stays uncluttered.

The lower section of the layout contains more specialized learning and collaboration spaces. On the left, the Embedded Lab is designed for hands-on technical work, offering multiple lab desks arranged for group or individual practical activities such as embedded systems development or electronics projects. Connected to this zone is a dedicated Video Conferencing Room, featuring a central conference table and equipment that supports remote meetings, presentations, and hybrid communication. To the right of this room, the Hybrid Room provides a flexible teaching

environment with neatly aligned desks and a smartboard, suitable for blended learning, seminars, or lessons requiring digital integration. Finally, the Washroom area occupies the far right end of the plan, equipped with multiple basins, several toilet stalls, and a designated accessible cubicle to support inclusivity and convenience for all users.

Overall, this floor design emphasizes efficiency, accessibility, and a smooth flow between learning, collaboration, and support spaces. Each room is arranged to enhance practicality—ranging from technical labs to meeting facilities—while essential services like storage, printing, and washrooms are strategically positioned to support daily academic operations.

3.0 APPENDIX

3.1 Projected Marks

RUBRICS	MARKS (10%)	PROJECTED MARKS
1. Group name and members	2	2
2. Suggested floor plan - Fit description - 2 general purpose lab (1 Cisco Network lab and 1 embedded lab) - 1 video conferencing rooms - 1 student lounge - 1 hybrid classroom - A total 30 workstations and other equipment planned for each lab - Scale: with scale - Clear and readable. Has appropriate labels - Creativity	8	8

3.2 Task Distribution

Name	Task Distribution
Tey Xin Yi	☒ Draft Floor Plan ☒ Design Ground Floor Plan
Lee Hui Zhen	☒ Draft Floor Plan ☒ Design First Floor Plan
Tan Hui Shan	☒ Task Introduction ☒ Draft Floor Plan ☒ Prepare Minutes Meeting

3.3 Meeting Minutes

MEETING MINUTES

DATE/TIME	15 November 2025 10:00AM		
LOCATION	BK2, N28		
AGENDA	1. Discuss group name 2. Way to arrange the floor plan 3. Tools for designing the floor plan 4. Task distribution for each person		
Meeting MC	Tey Xin Yi		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Tey Xin Yi	10:00		
Lee Hui Zhen	10:00		
Tan Hui Shan	10:00		
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss group name	- Lee suggest TTL Network Solutions (means “Time To Live” in networking) - All members agree with Lee’s suggestion	Lee (15/11)
2	Way to arrange the floor plan	- All members discussed the example floor plan that was provided by senior - Tan highlight the required rooms (general purpose labs, cisco networking lab, embedded lab, hybrid classroom, video conference room and student lounge) - Tey distributed rooms into 2 floors - Lee adds on essential additional rooms (IDF room, electrical room and storage room) - All members agreed with the suggested plan - All members discussed the size of the building - All members draft on the floor plan	All members (15/11)

3	Tools for designing the floor plan	• Tan suggests examples tools to use (Autocad, Draw.io, Floorplanner) • All members make comparisons between these tools to determine the most suitable tool for designing our floor plan. • All members agree to use Autocad for designing the floor plan	Tan (15/11)
4	Task distribution for each person	• Tey distributes tasks into 3 parts: 1) Design ground floor plan 2) Design first floor plan 3) Prepare whole proposal • All members choose on their own: ○ Tey-design ground floor plan ○ Lee-design first floor plan ○ Tan-prepare whole proposal	Tey (15/11)
5	Meeting ended	12:00 pm	

3.4 Meeting Photo

